

Machine Learning

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Definition & History

ML Types

Labeled vs Unlabeled

Regression vs Classification

Train/Val/Test

Confusion Matrix

AUC & ROC

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Definition & History of Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to **learn from data** and improve their performance on tasks **without being explicitly programmed** for every scenario.

Arthur Samuel (1959) first coined the term — he defined ML as *"the field of study that gives computers the ability to learn without being explicitly programmed."*

Key Definitions

Term	Definition / Contribution
Arthur Samuel (1959)	Coined the term ML — learning without explicit programming
Tom Mitchell (1997)	A program LEARNS from experience E w.r.t. task T if performance P improves
Deep Learning	ML using multi-layered neural networks for complex pattern recognition
Feature	An individual measurable property or characteristic of the data
Label / Target	The output variable the model is trying to predict

Brief History Timeline

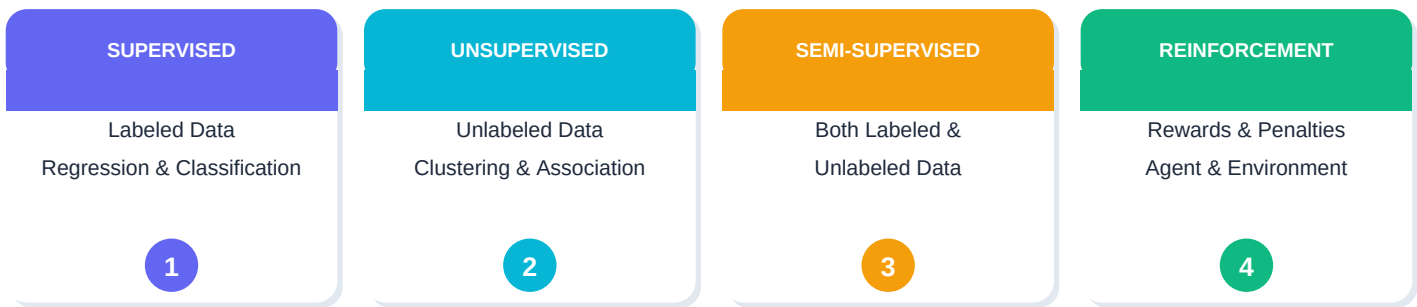
1943	McCulloch & Pitts propose artificial neuron model
1950	Alan Turing introduces the Turing Test
1957	Frank Rosenblatt creates the Perceptron
1959	Arthur Samuel coins the term 'Machine Learning'
1986	Rumelhart & Hinton popularize Backpropagation
1997	IBM Deep Blue defeats chess champion Kasparov
2006	Hinton's Deep Belief Networks — Deep Learning era begins
2012	AlexNet wins ImageNet — CNN revolution
2017	Transformer architecture introduced (Attention is All You Need)
2022+	LLMs (ChatGPT, Claude) — Generative AI era

Applications of Machine Learning

Domain	Applications
Healthcare	Disease diagnosis, drug discovery, medical imaging (X-ray, MRI analysis)
Finance	Fraud detection, credit scoring, algorithmic trading, risk assessment
NLP	Chatbots, translation, sentiment analysis, text summarization
Computer Vision	Face recognition, object detection, autonomous vehicles
Recommendation	Netflix, YouTube, Amazon product suggestions
Robotics	Path planning, manipulation, reinforcement learning in robots

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Types of Machine Learning



1. Supervised Learning

The algorithm learns from a **labeled dataset** — each input has a corresponding **known output (label)**. The model learns the mapping $f(X) \rightarrow Y$ and generalizes to unseen data.

- Training data: (X, Y) pairs — features + labels
- Goal: Learn f such that $f(X) \approx Y$
- Output: Continuous (Regression) or Discrete (Classification)
- Examples: Linear Regression, Decision Trees, SVM, Neural Networks
- Use Cases: Spam detection, price prediction, medical diagnosis

2. Unsupervised Learning

The algorithm is given **unlabeled data** — no explicit output is provided. The model must discover **hidden patterns, structures, or groupings** on its own.

- Training data: X only — no labels
- Goal: Discover hidden structure in data
- Types: Clustering (k-means), Dimensionality Reduction (PCA), Association Rules
- Examples: K-Means, DBSCAN, Autoencoders, Apriori
- Use Cases: Customer segmentation, anomaly detection, market basket analysis

3. Semi-Supervised Learning

Uses a **small amount of labeled data** combined with a **large amount of unlabeled data**. Bridges the gap between supervised and unsupervised learning — practical when labeling is expensive.

- Training data: Few (X,Y) pairs + many X-only samples
- Goal: Leverage unlabeled data to improve model accuracy
- Examples: Self-training, Co-training, Label Propagation
- Use Cases: Web content classification, medical imaging with few labeled scans
- Advantage: Reduces need for expensive human annotation

4. Reinforcement Learning

An **Agent** interacts with an **Environment** by taking **Actions** and receiving **Rewards or Penalties**. The agent learns a **Policy** to maximize cumulative reward over time.

Agent	The learner / decision maker
Environment	The world the agent interacts with
State (S)	Current situation of the agent
Action (A)	Choices the agent can make
Reward (R)	Feedback signal (positive or negative)
Policy (π)	Strategy: state $\rightarrow$ action mapping

- Examples: Q-Learning, Deep Q-Network (DQN), PPO, A3C
- Use Cases: Game playing (AlphaGo, Chess), robot navigation, self-driving cars
- Key equation: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$ (Bellman Equation)

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Labeled vs Unlabeled Datasets

Labeled Dataset	Unlabeled Dataset
Has input-output pairs (X, Y)	Has only input data (X)
Requires human annotation	No annotation required
More expensive to create	Easy & cheap to collect
Used in Supervised Learning	Used in Unsupervised Learning
Example: Email (spam/not spam)	Example: Raw text corpus
Smaller in practice	Typically larger in volume
Higher quality guarantee	Requires model to find patterns

Example: In a medical imaging dataset — an **X-ray image is the input (X)** and the **diagnosis label (cancer/no-cancer) is Y**. Obtaining Y requires expert radiologists, making labeled data expensive. Hospitals have millions of **unlabeled** X-rays but only thousands of **labeled** ones — hence Semi-Supervised Learning is valuable.

Types of Annotation / Labels

Type	Example Value	Use Case
Binary Labels	Yes/No, 0/1, True/False	Spam detection
Multi-class Labels	Cat/Dog/Bird/Fish	Image classification
Continuous Labels	House price: ■45 lakhs	Regression tasks
Sequence Labels	POS tags: Noun/Verb/Adj	NLP tasks (NER, POS)
Bounding Boxes	x,y,w,h coordinates	Object detection

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Supervised Learning Tasks: Regression vs Classification

REGRESSION	CLASSIFICATION
Predicts continuous values	Predicts discrete class labels
Output: real number (e.g. 45.7)	Output: category (e.g. 'Cat')
Evaluated by MSE, RMSE, R ²	Evaluated by Accuracy, F1, AUC
Linear Regression, SVR, GPR	Logistic Regression, SVM, DT
House price prediction	Email spam detection
Stock price forecasting	Image classification
Temperature prediction	Medical diagnosis (cancer/no)

Types of Regression

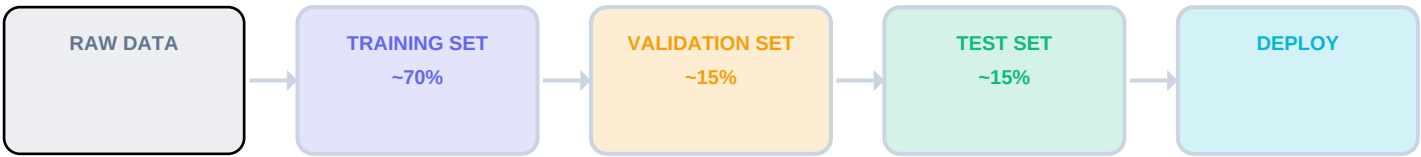
Type	Equation	Notes
Simple Linear Regression	$y = mx + c$	One feature → one output
Multiple Linear Regression	$y = w_1x_1 + w_2x_2 + \dots + b$	Multiple features → one output
Polynomial Regression	$y = w_0 + w_1x + w_2x^2 + \dots$	Non-linear relationships
Logistic Regression*	$\sigma(wx+b) \rightarrow P(y=1 x)$	*Actually Classification despite name
Ridge / Lasso	Linear + L2/L1 penalty	Regularized regression

Types of Classification

Type	Description	Example	Algorithm
Binary	2 classes only	Spam vs Not Spam	Logistic Regression, SVM
Multi-class	3+ mutually exclusive classes	Digit recognition (0-9)	Softmax, Decision Trees
Multi-label	Multiple labels per sample	Movie genres	Neural Networks
Imbalanced	Unequal class distribution	Fraud (rare event)	SMOTE, weighted loss

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Learning Framework: Training, Validation & Testing



Training Set	60–80%	Model learns parameters (weights)	Gradient descent, backpropagation
Validation Set	10–20%	Tune hyperparameters, prevent overfitting	Early stopping, model selection
Test Set	10–20%	Final unbiased evaluation — touched ONCE	Report final performance metrics

K-Fold Cross Validation

When data is limited, **K-Fold Cross Validation** gives a more reliable estimate of model performance by rotating the validation set:

- Shuffle dataset and split into K equal folds (typically K=5 or K=10)
- Train on K-1 folds, validate on the remaining 1 fold
- Repeat K times — each fold serves as validation exactly once
- Final performance = average metric across all K iterations
- Stratified K-Fold: maintains class distribution in each fold

Bias-Variance Trade-off

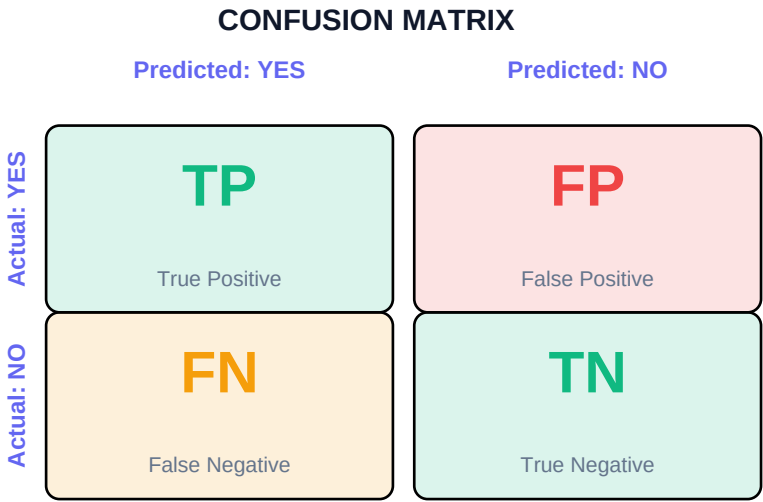
Underfitting (High Bias)	High	High	Model too simple; fails to capture patterns	More complex model, more features
Good Fit	Low	Low	Model generalizes well; ideal scenario	—
Overfitting (High Variance)	Very Low	High	Model memorizes training data; fails on new data	Regularization, more data, dropout

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Performance Evaluation Parameters

Confusion Matrix

A **Confusion Matrix** is a table that summarizes the performance of a classification model by comparing **Actual vs Predicted** labels.



TP — True Positive	Actual: Positive, Predicted: Positive ✓ Correct
TN — True Negative	Actual: Negative, Predicted: Negative ✓ Correct
FP — False Positive	Actual: Negative, Predicted: Positive ✗ Type I Error
FN — False Negative	Actual: Positive, Predicted: Negative ✗ Type II Error

Evaluation Metrics with Formulas

ACCURACY

$$\frac{TP+TN}{TP+TN+FP+FN}$$

Overall correct predictions

PRECISION

$$\frac{TP}{TP + FP}$$

Among +ve predicted, truly +ve

RECALL

$$\frac{TP}{TP + FN}$$

Among actual +ve, caught correctly

F1 SCORE

$$2 \times \frac{P \times R}{P + R}$$

Harmonic mean of P & R

SPECIFICITY

$$\frac{TN}{TN + FP}$$

True Negative Rate

Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	% of total correct predictions. Misleading for imbalanced data.	Overall correctness
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Precision	$TP/(TP+FP)$	Of all predicted positives, how many are truly positive? Important when FP is costly.	Anti-spam: avoid flagging valid emails
Recall (Sensitivity)	$TP/(TP+FN)$	Of all actual positives, how many did we catch? Important when FN is costly.	Cancer diagnosis: must not miss cases
F1 Score	$2 \cdot (P \cdot R) / (P + R)$	Harmonic mean of Precision & Recall. Use when both FP and FN matter.	Information retrieval, NLP
Specificity (TNR)	$TN/(TN+FP)$	Of all actual negatives, how many correctly identified? Complement of FPR.	Medical tests — ruling out disease
AUC-ROC	Area under ROC curve	AUC=1: perfect, AUC=0.5: random. Threshold-independent metric.	Binary classifier comparison

ROC Curve & AUC

ROC (Receiver Operating Characteristic) curve plots **True Positive Rate (Recall)** vs **False Positive Rate (1–Specificity)** at various classification thresholds.

- X-axis: $FPR = FP / (FP + TN) \rightarrow$ False Alarm Rate
- Y-axis: $TPR = TP / (TP + FN) \rightarrow$ Sensitivity / Recall
- Perfect Classifier: $AUC = 1.0$ (curve hugs top-left corner)
- Random Classifier: $AUC = 0.5$ (diagonal line)
- $AUC > 0.9$: Excellent | $0.8\text{--}0.9$: Good | $0.7\text{--}0.8$: Fair | <0.7 : Poor
- AUC is threshold-independent — great for comparing classifiers

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Quick Revision — Key Formulas & Definitions

Formula Cheat Sheet

Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision (PPV)	$TP / (TP + FP)$
Recall (Sensitivity / TPR)	$TP / (TP + FN)$
Specificity (TNR)	$TN / (TN + FP)$
False Positive Rate (FPR)	$FP / (FP + TN) = 1 - \text{Specificity}$
F1 Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$
F-beta Score	$((1+\beta^2) \times P \times R) / (\beta^2 \times P + R)$ [$\beta=2 \rightarrow$ Recall favored, $\beta=0.5 \rightarrow$ Precision]
MSE (Regression)	$\Sigma(y_{\text{actual}} - y_{\text{predicted}})^2 / n$
RMSE	$\sqrt{\text{MSE}}$
R ² Score	$1 - (SS_{\text{res}} / SS_{\text{tot}})$ [1=perfect, 0=baseline]

One-Liner Definitions (Exam Ready)

Term	Definition
Machine Learning	Algorithms that learn from data to make predictions/decisions without explicit programming
Supervised Learning	Learning with labeled data; model learns input→output mapping
Unsupervised Learning	Learning without labels; finds hidden patterns/clusters in data
Semi-Supervised	Uses small labeled + large unlabeled dataset for learning
Reinforcement Learning	Agent learns optimal policy by receiving rewards/penalties from environment
Overfitting	Model performs well on training data but poorly on unseen data — memorizes noise
Underfitting	Model is too simple — poor performance on both train and test data
Bias	Error from wrong assumptions — leads to underfitting
Variance	Sensitivity to small fluctuations — leads to overfitting
Cross-Validation	Technique to estimate model performance by training on multiple data splits
Confusion Matrix	Table comparing actual vs predicted class labels — 4 cells: TP,TN,FP,FN
AUC	Area under ROC curve — measures classifier's ability to distinguish classes
Precision	Of predicted positives, how many are truly positive (reduces false alarms)
Recall	Of actual positives, how many did the model detect (reduces missed cases)
F1 Score	Harmonic mean of Precision and Recall — balanced metric

EXAM TIPS — COMMON MISTAKES TO AVOID

- Logistic Regression IS a Classification algorithm (not regression)
- Accuracy is misleading for imbalanced datasets — prefer F1 / AUC
- FP = Type I Error | FN = Type II Error
- Test set must NEVER be used for hyperparameter tuning — use Validation
- AUC = 0.5 means model is no better than random guessing
- Recall = Sensitivity = TPR (same metric, different names)
- High Precision + Low Recall: model is conservative (misses many positives)
- High Recall + Low Precision: model is liberal (flags too many false positives)